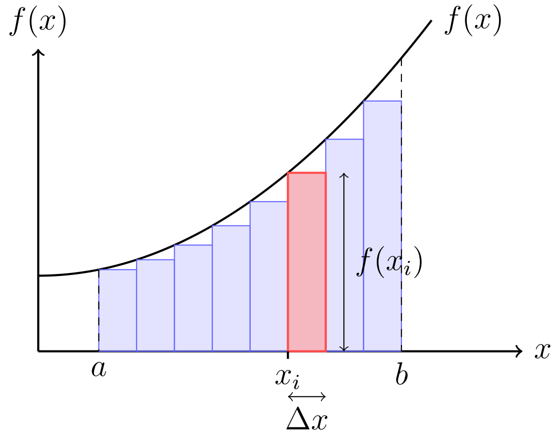
Topic 8: Integrals[1]

Fundamental Theorem of Calculus



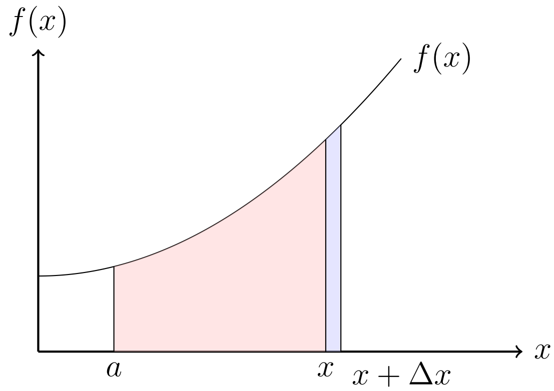
Definition 1 (Integral)

let a function $f : \mathbb{R} \to \mathbb{R}$ be continuous in the close interval $a \leq x \leq b$. We divide the interval by $(n-1)$ points, such that $x_i - x_{i-1} = \Delta x$, where $\Delta x = \frac{b-a}{n}$:

$$\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_{i=1}^{n} f(x_i)\Delta x$$

Denote the integral of a function f from a to x as $F(x)$: $F(x) = \int_a^x(t)dt$. From the diagram below, observe that when x increases by a small amount Δx, the change in the accumulated area, given by $F(x+\Delta x) - F(x)$, is approximately equal to the function value at x.

[1] Instructors: Yue Yu and Sofia Olguin. This note was prepared for the 2026 UCSB Math Camp for Ph.D. students in economics. It incorporates materials from previous instructors, including Shu-Chen Tsao, ChienHsun Lin, and Sarah Robinson.



Theorem 1: First Fundamental Theorem of Calculus

Let f be a Riemann integrable function on $[a, b]$. For $a \le x \le b$, put

$$F(x) = \int_a^x f(t)\ dt.$$

Then, F is (uniformly) continuous on $[a, b]$.

Furthermore, if f is continuous at a point x_0 of $[a, b]$, then F is differentiable at x_0, and

$$F'(x_0) = f(x_0).$$

The First Fundamental Theorem of Calculus suggests that integration and differentiation are inverse operations. For this reason, the integral of f is sometimes called the **antiderivative** of f.

Theorem 2: Second Fundamental Theorem of Calculus

Let f be a Riemann integrable function on $[a, b]$. Suppose there is a differentiable function F on $[a, b]$ such that $F' = f$, then

$$\int_a^b f(x)\ dx = F(b) - F(a).$$

Properties of Integrals

The following are some rules for integrals. Assume all functions are Riemann integrable.

Proposition 1

1. Linearity: $\int_a^b f(x) + g(x)dx = \int_a^b f(x)dx + \int_a^b g(x)dx$.

2. Bounds: Let m be a lower bound of $f(x)$ for all $x \in [a, b]$, and let M be an upper bound of $f(x)$ for all $x \in [a, b]$. Then, $m(b - a) \leq \int_a^b f(x)dx \leq M(b - a)$.

3. Inequalities between functions: If $f(x) \leq g(x)$ for each $x \in [a, b]$, then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$.

Example 1

Suppose a company expects to receive a continuous income stream at a rate of $R(t) = 2000e^{0.05t}$ dollars per year, where t is the time in years.

Calculate the present value of this income stream over the next 10 years if the discount rate is 2 percent.

One useful theorem that we learned in freshmen calculus is the integration by parts.

Theorem 3: Integration by parts

Suppose F and G are differentiable on $[a, b]$, and f and g are Riemann integrable. Then,

$$\int_a^b F(x)g(x)\ dx = (F(b)G(b) - F(a)G(a)) - \int_a^b f(x)G(x)\ dx$$

Example 2

Evaluate the following integral

$$\int_a^b xe^x\ dx.$$

The first step for integration by parts is to determine the parts.

Let $F(x) = x$, and $G(x) = e^x$. Then, $f(x) = F'(x) = 1$, and $g(x) = g'(x) = e^x$.

Hence, we can replace our integral by

$$\begin{aligned}\int_a^b xe^x dx &= be^b - ae^a - \int_a^b e^x dx \\ &= be^b - ae^a - \left[e^x\right|_a^b \\ &= be^b - ae^a - \left(e^b - e^a\right) = (b-1)e^b - (a-1)e^a.\end{aligned}$$

Leibniz rule

Theorem 4

Suppose $f(x,t)$ is continuously differentiable with respect to t, and $a(t), b(t)$ are differentiable.[a]

Then,

$$\frac{d}{dt}\left(\int_{a(t)}^{b(t)} f(x,t)\,dx\right) = \int_{a(t)}^{b(t)} \frac{\partial}{\partial t} f(x,t)\,dx + f(b(t),t)\cdot b'(t) - f(a(t),t)\cdot a'(t)$$

[a] Continuously differentiable: the derivative exists and is continuous.

Example 3

Evaluate the following integral

$$\frac{d}{dt}\int_0^{t^2} e^{xt}\,dx.$$

Useful inequalities for integrals

This section revisits four useful inequalities: Jensen's, Cauchy-Schwarz, Hölder's, and Minkowski. Although we originally defined them in algebraic or metric space settings, each of these inequalities extends naturally to integrals.

These inequalities appear frequently in applications during the first year, particularly for econometrics courses.

Theorem 5: Jensen's Inequality in $\mathbb{R}^n$

Consider any $\mathbf{x}_1, ..., \mathbf{x}_n \in \mathbb{R}^n$, and any $\lambda_1, ..., \lambda_n \in [0,1]$ such that $\sum_{i=1}^n \lambda_i = 1$. This forms a convex combination $\sum_{i=1}^n \lambda_i \mathbf{x}_i$. Suppose f is convex.

Jensen's inequality states:

$$f\left(\sum_{i=1}^n \lambda_i \mathbf{x}_i\right) \leq \sum_{i=1}^n \lambda_i f(\mathbf{x}_i)$$

Proposition 2 (Jensen's Inequality in Integral Form)

Let g be a convex function and f a measurable function such that both $g(f(x))$ and $f(x)$ are integrable.

Jensen's inequality states:

$$g\left(\int_X f(x)\,d\mu(x)\right) \leq \int_X g(f(x))\,d\mu(x).$$

where $\mu(x)$ is a measure (e.g., probability measure) on the space X.

The statistical version of Jensen's inequality is

$$h(\mathbb{E}[X]) \leq \mathbb{E}[h(X)]$$

where h is a convex function, X is a random variable, and $\mathbb{E}[\cdot]$ is the expectation.

Theorem 6: Cauchy-Schwarz Inequality in $\mathbb{R}^n$

Consider any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, where $\mathbf{x} = (x_1, ..., x_n)$ and $\mathbf{y} = (y_1, ..., y_n)$. The Cauchy-Schwarz inequality states:

$$\left(\sum_{i=1}^n x_i y_i\right)^2 \leq \left(\sum_{i=1}^n x_i^2\right)\left(\sum_{i=1}^n y_i^2\right)$$

Proposition 3 (Cauchy-Schwarz Inequality in Integral Form)

The Cauchy-Schwarz inequality states:

$$\left(\int_X f(x)g(x)\,d\mu(x)\right)^2 \leq \left(\int_X |f(x)|^2\,d\mu(x)\right)\left(\int_X |g(x)|^2\,d\mu(x)\right).$$

The statistical version of the Cauchy-Schwarz inequality is

$$(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$$

Theorem 7: Hölder's Inequality in $\mathbb{R}^n$

Consider any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, where $\mathbf{x} = (x_1, ..., x_n)$ and $\mathbf{y} = (y_1, ..., y_n)$. The Hölder's inequality states:

$$\sum_{i=1}^{n} |x_i y_i| \leq \left(\sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}} \left(\sum_{i=1}^{n} |y_i|^q \right)^{\frac{1}{q}}$$

for all $p, q \in [1, \infty)$ and $\frac{1}{p} + \frac{1}{q} = 1$.

Proposition 4 (Hölder's Inequality Inequality in Integral Form)

Hölder's inequality states:

$$\int_X |f(x)g(x)| \, d\mu(x) \leq \left(\int_X |f(x)|^p \, d\mu(x) \right)^{1/p} \left(\int_X |g(x)|^q \, d\mu(x) \right)^{1/q}.$$

for all $p, q \in [1, \infty)$ and $\frac{1}{p} + \frac{1}{q} = 1$.

The statistical version of Hölder's inequality is

$$\mathbb{E}[|XY|] \leq \mathbb{E}[|X|^p]^{\frac{1}{p}} \mathbb{E}[|Y|^q]^{\frac{1}{q}}$$

Theorem 8: Minkowski inequality $\mathbb{R}^n$

Consider any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, where $\mathbf{x} = (x_1, ..., x_n)$ and $\mathbf{y} = (y_1, ..., y_n)$. The Minkowski inequality states:

$$\left(\sum_{i=1}^{n} |x_i + y_i|^p \right)^{\frac{1}{p}} \leq \left(\sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}} + \left(\sum_{i=1}^{n} |y_i|^p \right)^{\frac{1}{p}} \quad \text{for all } p \in [1, \infty).$$

Proposition 5 (Minkowski inequality in Integral Form)

The Minkowski inequality states:

$$\left(\int_X |f(x) + g(x)|^p \, d\mu(x) \right)^{1/p} \leq \left(\int_X |f(x)|^p \, d\mu(x) \right)^{1/p} + \left(\int_X |g(x)|^p \, d\mu(x) \right)^{1/p}.$$

The statistical version of the Minkowski inequality is

$$(\mathbb{E}[|X + Y|^p])^{\frac{1}{p}} \leq \mathbb{E}[|X|^p]^{\frac{1}{p}} \mathbb{E}[|Y|^p]^{\frac{1}{p}}$$